生物统计和生物信息引论

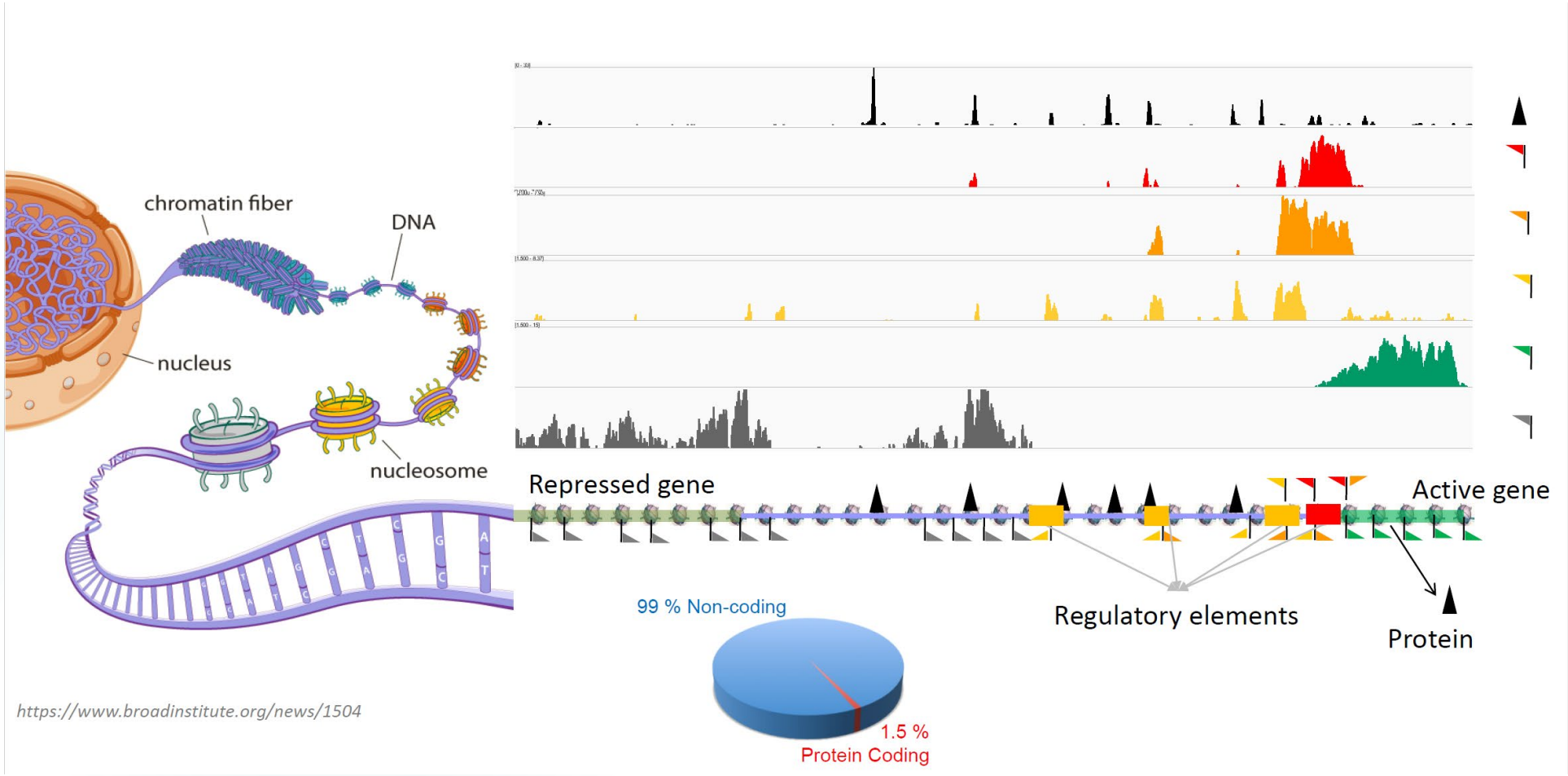
Lecture 11

生物医学问题中的复杂模型应用

王健桥
jianqw@mail.tsinghua.edu.cn

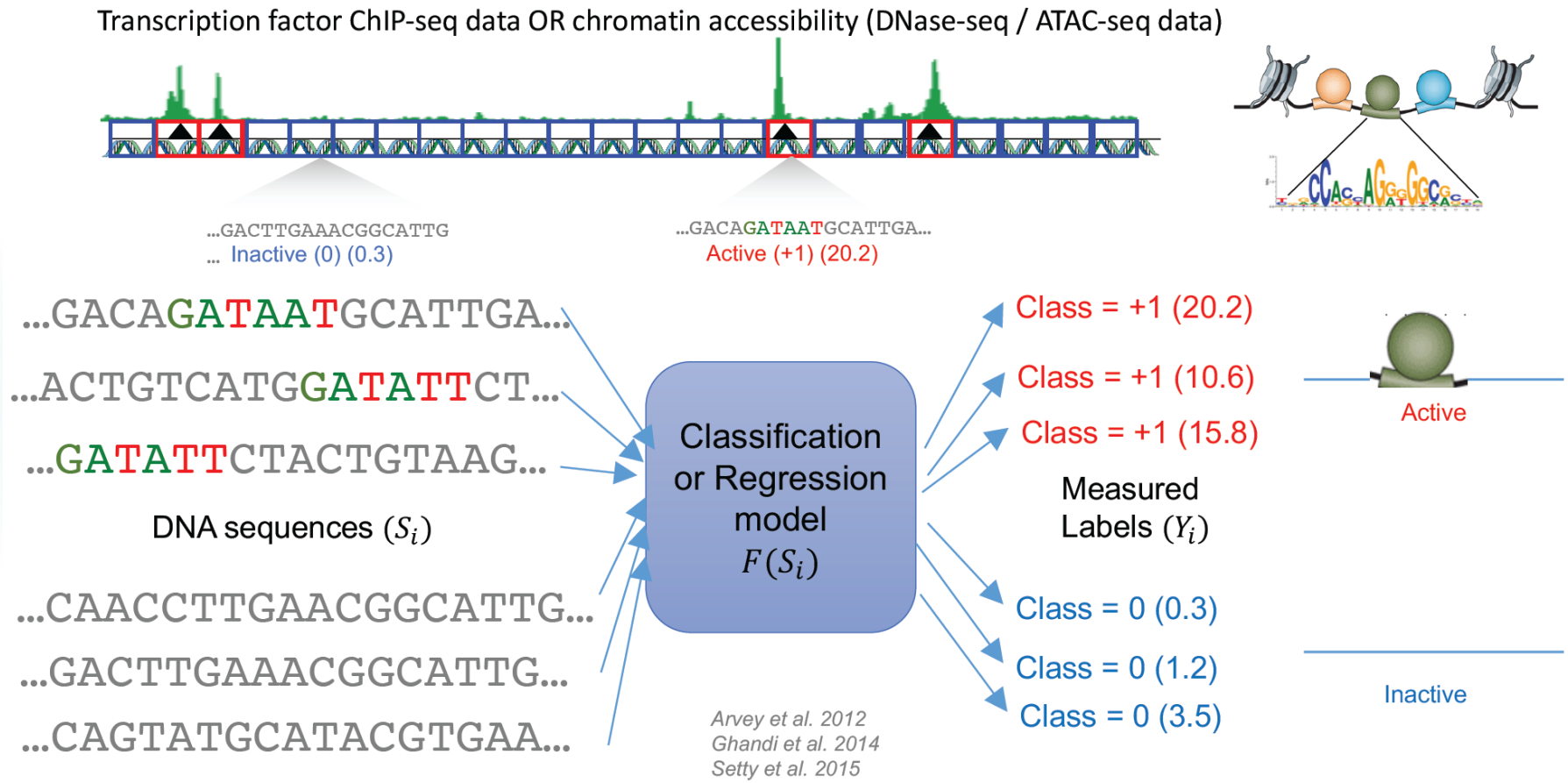
清華大學 统计与数据科学系
Department of Statistics and Data Science, Tsinghua University

From sequence to regulatory variants



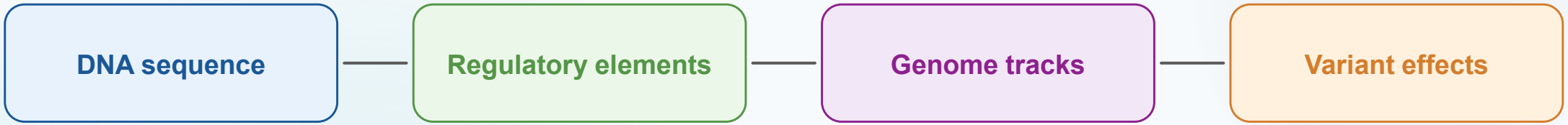
https://www.broadinstitute.org/news/1504

DNA model


Transcription factor ChIP-seq data OR chromatin accessibility (DNase-seq / ATAC-seq data)
...GACTTGAAACGGCATTG
... Inactive (0) (0.3)
...GACAGATAATGCATTGA...
Active (+1) (20.2)
...GACAGATAATGCATTGA...
...ACTGTCATGGATATTCT...
...GATATTCTACTGTAAG...
DNA sequences (S_i)
...CAACCTTGAACGGCATTG...
...GACTTGAAACGGCATTG...
...CAGTATGCATACGTGAA...
Classification or Regression model $F(S_i)$
Arvey et al. 2012
Ghandi et al. 2014
Setty et al. 2015
Class = +1 (20.2)
Class = +1 (10.6)
Class = +1 (15.8)
Measured Labels (Y_i)
Class = 0 (0.3)
Class = 0 (1.2)
Class = 0 (3.5)
Active
Inactive

AlphaGenome

From DNA sequence to regulatory function



Goal: understand how changes in DNA sequence alter gene regulation and disease risk

清華大學 统计与数据科学系
Department of Statistics and Data Science, Tsinghua University

Dataset: What Is the Supervised Learning Target?

Labels are experimental genomic measurements, not disease labels.

Data type	Example assay	Biological meaning
Expression	RNA-seq / CAGE / PRO-cap	How much RNA is produced; where transcription starts
Accessibility	ATAC-seq / DNase-seq	Which DNA regions are open and active
Chromatin state	Histone ChIP-seq / TF ChIP-seq	Marks and protein binding at regulatory regions
3D genome	Hi-C / Micro-C	Which genomic regions are close in space
Splicing	splice sites / junctions	How transcripts are processed

Training examples are genomic intervals paired with many tracks across cell types and tissues.

Roadmap for the AlphaGenome Section

We first motivate the problem; only then introduce the model.

1. Why is the problem important?

Non-coding variants and gene regulation

2. Biological background

Regulatory elements, genome tracks, cell-type specificity

3. Dataset and learning target

DNA sequence → experimental genomic assays

4. Applications

Variant effects, disease mechanisms, regulatory design

5. Model introduction

Input/output, architecture, training, benchmarks

Teaching design: emphasize biological problem formulation before model architecture

Why Is This Problem Important?

Protein-coding regions are only one layer of genome function.

Coding variants

May alter protein sequence

Often easier to reason about

Example: missense variant

Non-coding variants

May alter regulatory logic

Often cell-type specific

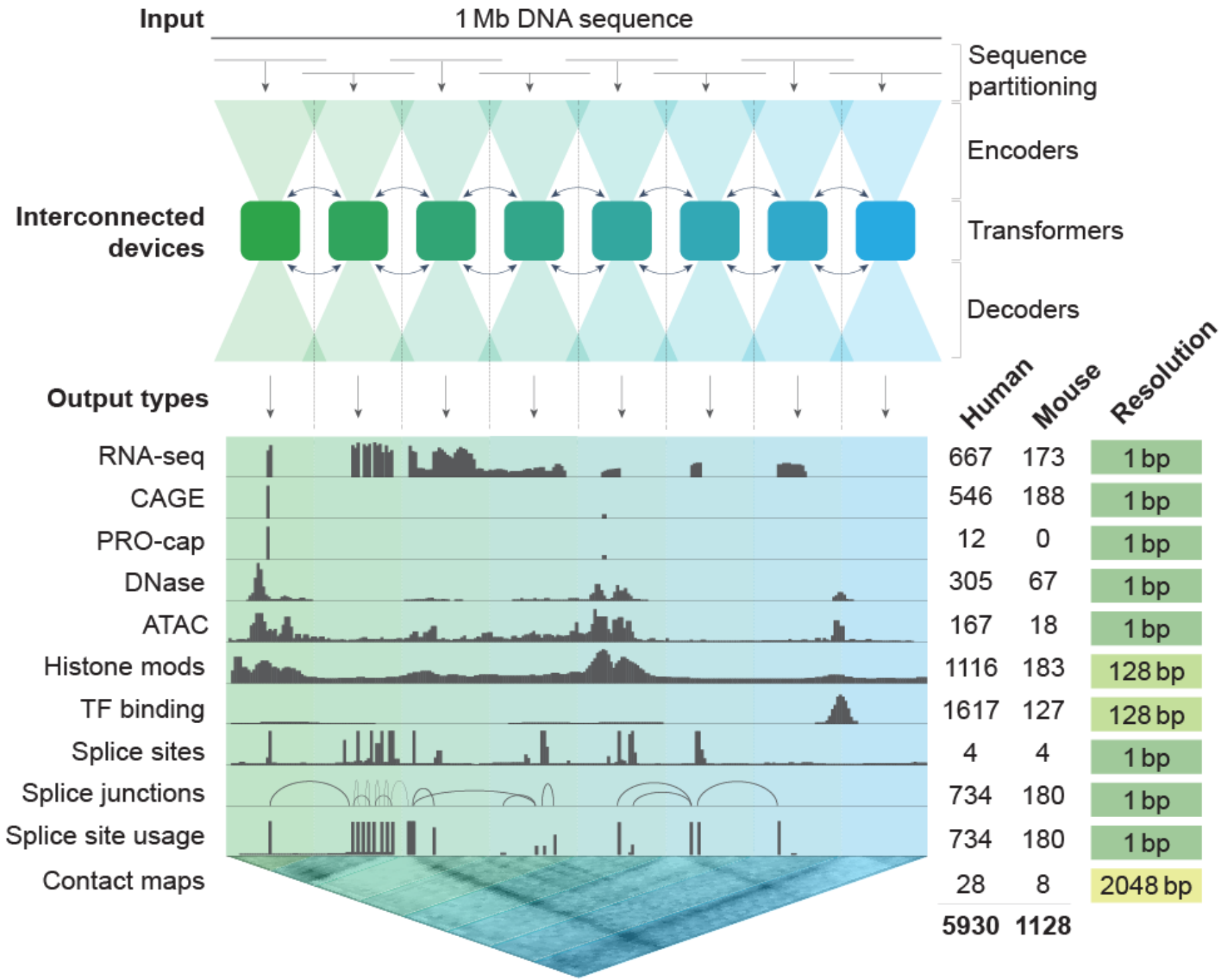
Harder to interpret mechanistically

Disease relevance

Variants may affect expression, splicing, chromatin accessibility, TF binding, or 3D contacts

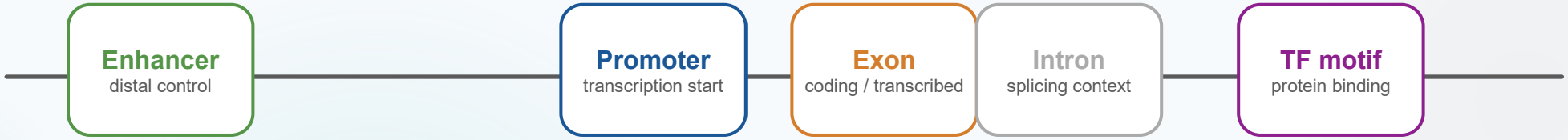
Core question

If one DNA base changes, what molecular process changes — and in which cell type?


Input
1 Mb DNA sequence
Sequence partitioning
Encoders
Interconnected devices
Transformers
Decoders
Output types
Human
Mouse
Resolution
RNA-seq 667 173 1 bp
CAGE 546 188 1 bp
PRO-cap 12 0 1 bp
DNase 305 67 1 bp
ATAC 167 18 1 bp
Histone mods 1116 183 128 bp
TF binding 1617 127 128 bp
Splice sites 4 4 1 bp
Splice junctions 734 180 1 bp
Splice site usage 734 180 1 bp
Contact maps 28 8 2048 bp
5930 1128

Background: The Regulatory Genome

The central dogma tells us information flow; regulation tells us when and where it happens.



Local signals

TF binding motifs, splice sites, polyadenylation signals

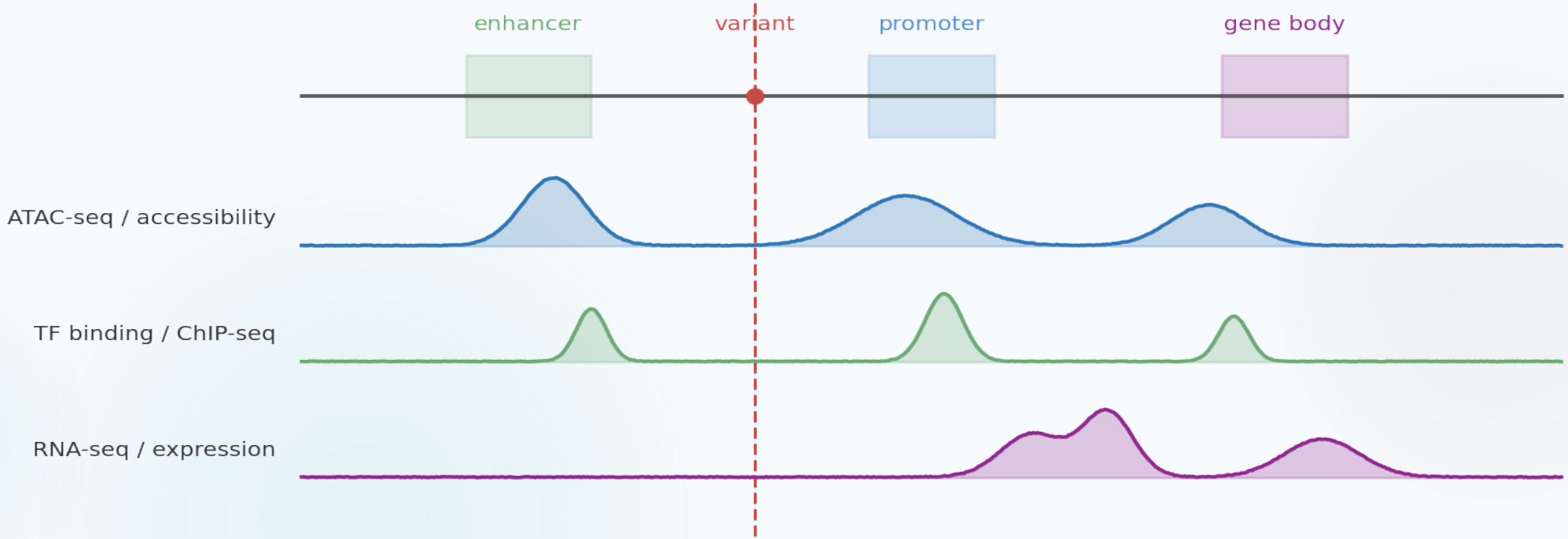
Long-range signals

Enhancer–promoter interactions and chromatin contacts

Biological background for sequence-to-function models; regulatory elements can act locally or distally

What Does the Output Look Like? Genome Tracks

A genome track assigns a signal value to positions along DNA.



Input

DNA sequence around a genomic region

Output

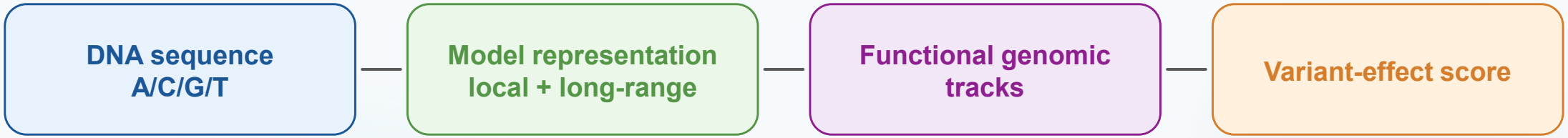
Predicted assay signal at many positions

Interpretation

Where regulation is active or changed

Problem Formulation: Sequence-to-Function

AlphaGenome is not a text-generation model; it predicts biological measurements.



Mathematically
:

f(sequence, context) → expression, accessibility, TF binding, splicing, contact maps, ...

LLMs learn contextual representations of words; AlphaGenome learns contextual representations of DNA regulatory sequence.

Dataset Sources and Scale

AlphaGenome learns from public functional genomics data.

ENCODE
regulatory genomics
chromatin, TF binding, expression

GTEx
tissue expression
human transcriptomics

4D Nucleome
3D genome organization
chromatin contacts

FANTOM5
transcription initiation
CAGE-based expression

Input scale
Up to 1 Mb DNA sequence
Human or mouse species identity

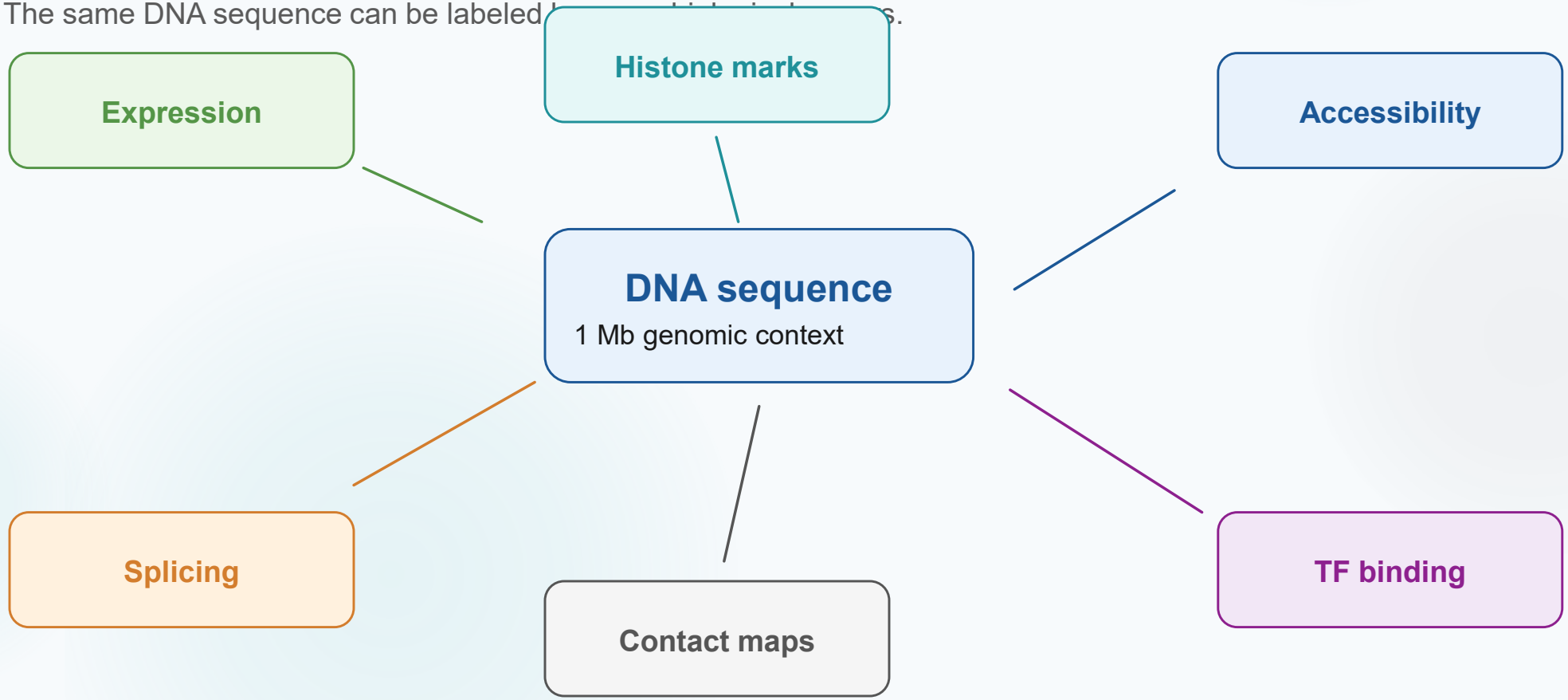
Output scale
5,930 human tracks
1,128 mouse tracks
11 output types

Key dataset idea
One input sequence is paired with many experimental readouts across biological contexts.

Source: Avsec et al., Nature 2026; Google DeepMind AlphaGenome blog: ENCODE, GTEx, 4D Nucleome, FANTOM5

Why the Dataset Is Multimodal

The same DNA sequence can be labeled by many biological processes.



A unified model can learn shared regulatory representations across modalities.

Why Existing Models Were Limited

AlphaGenome targets two major trade-offs.

Trade-off 1: long context vs high resolution

Short-window models can predict base-resolution features but may miss distal elements. Long-context models capture broad context but often reduce output resolution.

Trade-off 2: generality vs specialization

Specialized models perform well for one modality; multimodal models can cover many readouts but may lag on specialized tasks.

AlphaGenome design goal

1 Mb context — **base-pair resolution** — **multi-modal outputs** — **variant scoring**

Source: Avsec et al., Nature 2026, discussion of sequence length/resolution and modality trade-offs

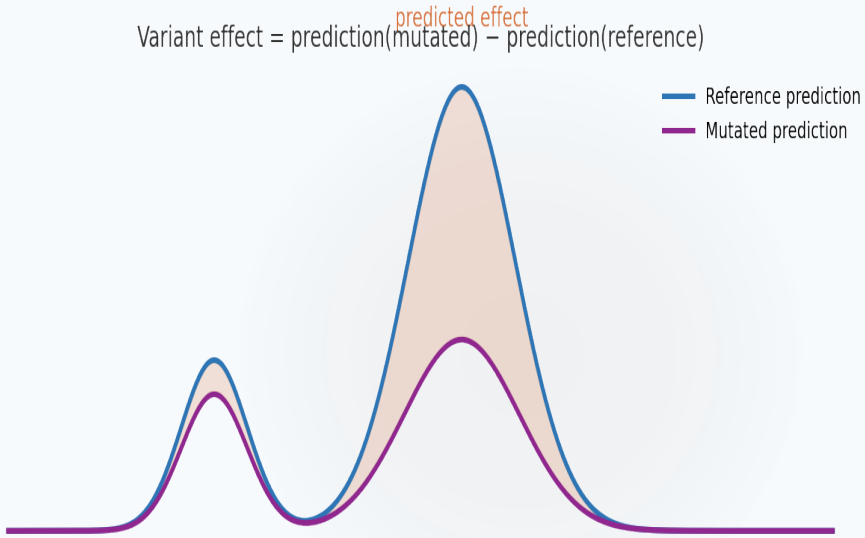
Application 1: Variant Effect Prediction

Compare predictions for reference and mutated sequences.

Reference sequence
ACGT...A...TGCA

Mutated sequence
ACGT...G...TGCA

AlphaGenome
same model

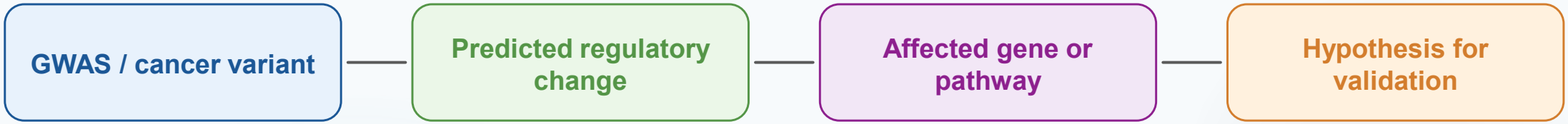


Interpretation

The difference between tracks suggests which molecular mechanism the variant may perturb.

Application 2: Disease Mechanism Discovery

Move from association to molecular mechanism.



- Prioritize candidate causal variants in non-coding regions
- Connect variants to expression, splicing, TF binding, accessibility or chromatin contacts
- Generate testable hypotheses for wet-lab validation

Example from paper/blog

T-ALL-associated mutations near TAL1: predicted introduction of a MYB binding motif and activation of TAL1.

Source: Google DeepMind blog; Avsec et al., Nature 2026, cross-modality variant interpretation near TAL1

Application 3: Beyond Variant Annotation

AlphaGenome can support several research directions.

Disease understanding

interpret non-coding variants

identify potential mechanisms

prioritize therapeutic targets

Synthetic biology

design regulatory DNA

cell-type-specific activation

optimize expression patterns

Fundamental genomics

map functional elements

study regulatory grammar

compare cell types/tissues

Important caution

Predictions are useful hypotheses for research, but experimental validation is still required.

Source: Google DeepMind blog, “A powerful research tool” and current limitations

AlphaGenome Paper: Key Contribution

A unified model for long-context, high-resolution, multimodal regulatory prediction.

Long context
Processes 1 Mb DNA sequence

High resolution
Predicts tracks up to single-base-pair resolution

Multimodal outputs
Expression, chromatin, splicing, contacts

Variant scoring
Reference vs mutated sequence

One sentence:

AlphaGenome predicts how DNA sequence encodes regulatory activity and how variants perturb that activity.

Source: Avsec et al., Nature 2026 abstract and main text

Finally: Introducing the Model

From biological problem to model input/output.

1 Mb DNA + species → **Sequence embeddings** → **Encoder local patterns** → **Transformer long-range context** → **Decoder + heads track predictions**

Convolutions

detect local motifs and fine-scale patterns

Transformers

model longer-range dependencies

Output heads

predict different modalities at appropriate resolutions

Source: Avsec et al., Nature 2026, Fig. 1 model overview; Google DeepMind architecture description

Model Architecture: What Each Part Does

Keep the explanation data-centric for undergraduates.

Component	Role in AlphaGenome	Biological intuition
Convolutional layers	Model local sequence patterns	motifs, splice sites, footprints
Transformer blocks	Exchange information across distant positions	enhancer–promoter interactions
U-Net-style encoder/decoder	Downsample for efficiency; upsample for resolution	long context + fine output
1D embeddings	Predict linear genome tracks	expression, accessibility, ChIP, splicing
2D embeddings	Predict pairwise contacts	3D chromatin interaction maps

Source: Avsec et al., Nature 2026, U-Net-inspired backbone, 1D/2D embeddings, convolutional and transformer blocks

Training: Pretraining and Distillation

The model learns experimental genomic tracks, then becomes efficient for variant scoring.

1. Pretraining

Sample 1-Mb genomic intervals

Train against experimental targets

Use cross-validation folds

2. Teacher models

Fold-specific and all-fold models

Strong track prediction performance

3. Student model

Distilled single model

Efficient variant effect prediction

Why distillation matters

For variant scoring, researchers want one model call that summarizes effects across many modalities and cell types.

Source: Avsec et al., Nature 2026, pretraining and distillation process

Evaluation: What Did the Paper Show?

Benchmarks test both track prediction and variant-effect prediction.

Genome track prediction

Evaluated on held-out genome regions

State-of-the-art on 22 of 24 track prediction tasks

Variant effect prediction

Matched or exceeded strongest external models in 25 of 26 evaluations

Interpretation for students

The main conceptual advance is not only accuracy. It is the unification of long context, high resolution and multimodal regulatory prediction in one model.

Source: Avsec et al., Nature 2026, abstract and main evaluation summary

Limitations and Scientific Caution

Predictions should be treated as hypotheses, not final biological truth.

- Very distant regulatory effects remain challenging.
- Cell- and tissue-specific regulation can still be difficult.
- The model predicts molecular outcomes, not full organismal disease risk.
- It has not been designed or validated for personal genome prediction or direct clinical decision-making.
- Experimental validation remains essential.

Classroom takeaway

AlphaGenome is a powerful research tool for hypothesis generation, not a replacement for biological experiments.

Source: Google DeepMind AlphaGenome blog, current limitations and research-use guidance

https://deepmind.google/blog/alphagenome-ai-for-better-understanding-the-genome/

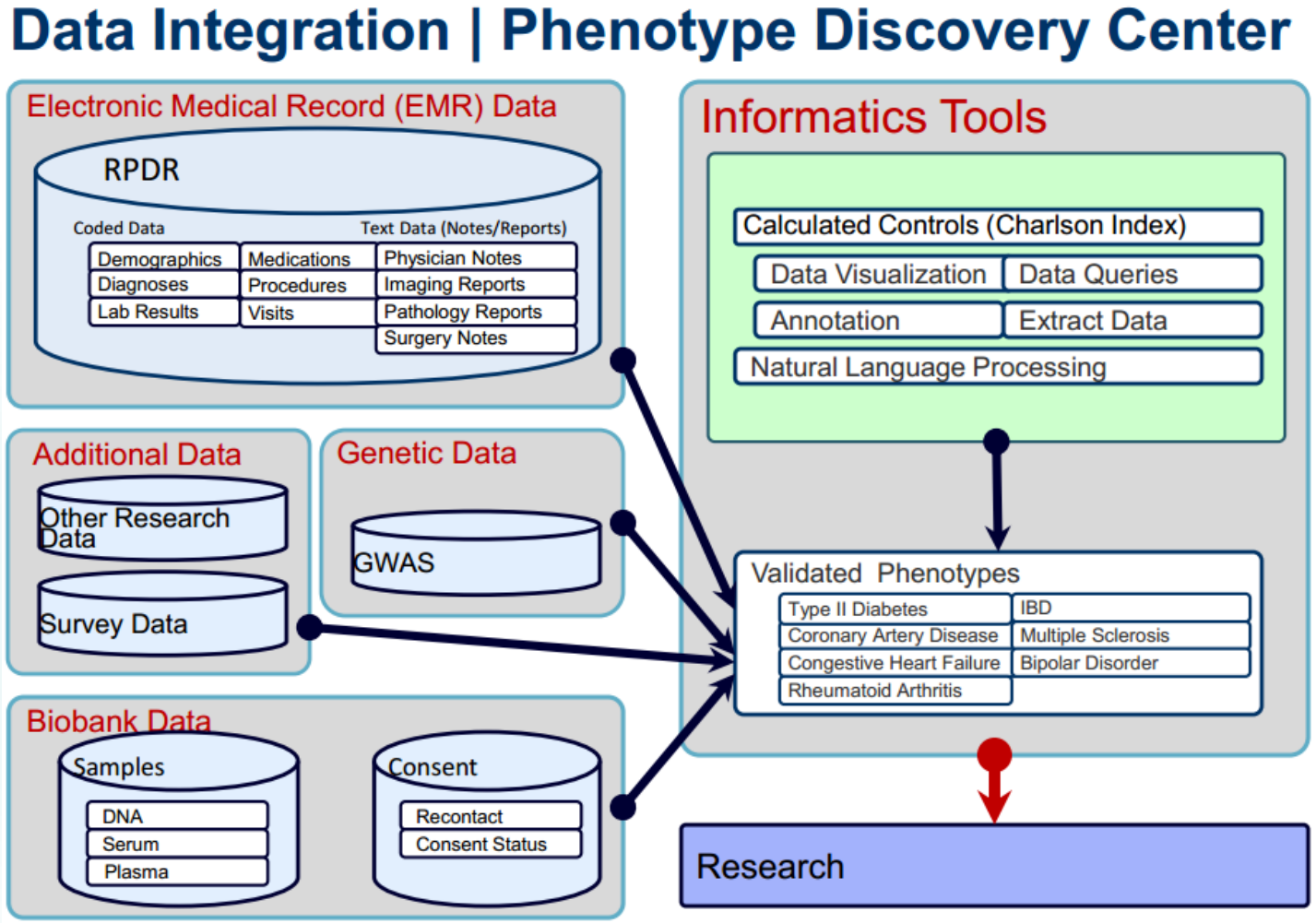
https://www.alphagenomedocs.com/tutorials/index.html

结构化让电子病历释放应用价值

- 原始应用
  - 临床流程电子化

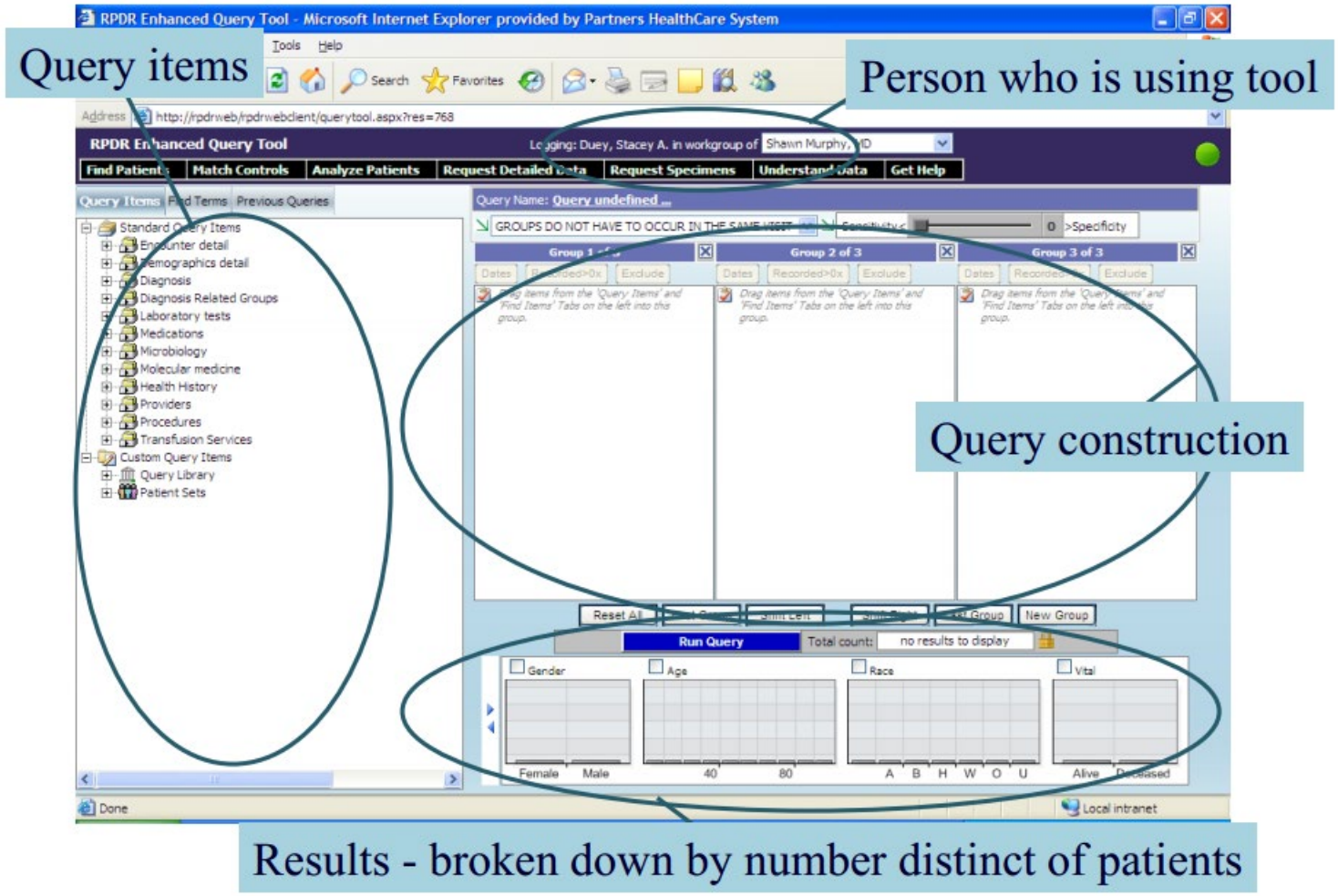
- 衍生应用
  - 临床科研数据
  - 政府监管数据
  - 政策制定（公共卫生政策、医保产品开发、医药市场研究等）
  - 临床决策支持模型

为科研服务的医疗数据集成与提取



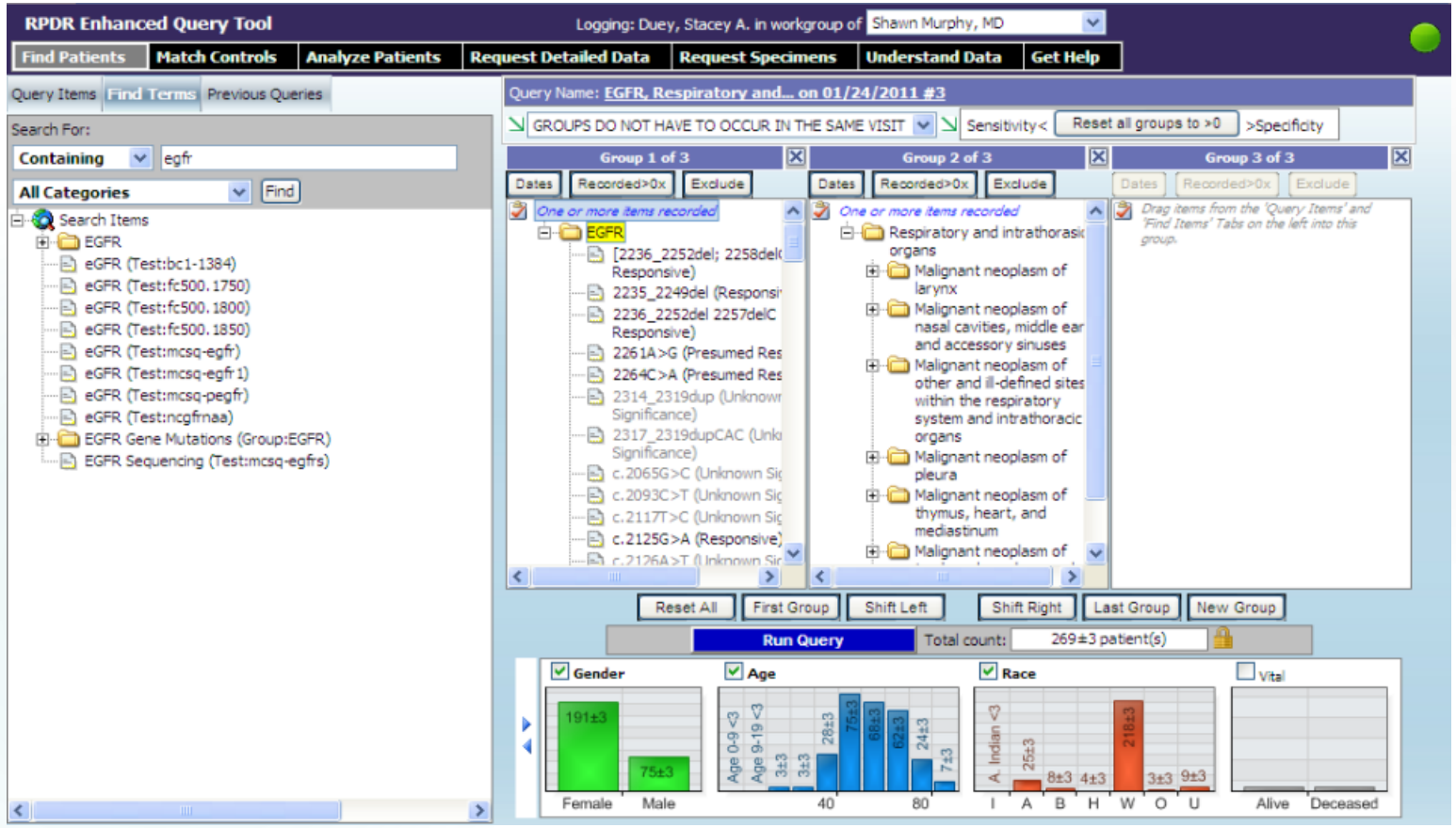
Source: Shawn Murphy, M.D., Ph.D., Corporate Director of Research Information Systems and Computing at Partners HealthCare

为科研服务的医疗数据集成与提取



Source: Shawn Murphy, M.D., Ph.D., Corporate Director of Research Information Systems and Computing at Partners HealthCare

为科研服务的医疗数据集成与提取



Source: Shawn Murphy, M.D., Ph.D., Corporate Director of Research Information Systems and Computing at Partners HealthCare

结构化让电子病历释放应用价值

- 原始应用
  - 临床流程电子化

- 衍生应用
  - 临床科研数据
  - 政府监管数据
  - 政策制定（公共卫生政策、医保产品开发、医药市场研究等）
  - 临床决策支持模型

- 多数关键信息蕴藏在自由文本中 —— 结构化是医疗信息有效利用的底层技术。

通用模型做结构化（DeepSeek r1）

下面是一篇脱敏的病历。提取患者服用的各种药物的剂量，以“药品名称：剂量”的格式逐行输出。慢性乙型肝炎病史10余年，曾有肝功能异常，中医治疗后好转；1年余前查HBsAg转阴，但肝脏病理提示病毒性肝炎伴肝纤维化（G1S3-4），予恩替卡韦0.5mg qd抗病毒治疗至今。2型糖尿病11年余，目前口服二甲双胍维格列汀850/50mg bid降糖治疗，血糖控制良好。血管性痴呆2年余，长期口服重酒石酸卡巴拉汀1.5mg bid。高血压1年余，血压最高180/80mmHg，长期口服硝苯地平控释片30mg qd，厄贝沙坦/氢氯噻嗪片150mg/12.5mg qn，血压控制在130/80mmHg。发现右侧锁骨下动脉起始部斑块形成1年余，未予特殊诊治。发现左侧肾上腺增粗、左侧肾盂输尿管连接处狭窄、继发左肾积水1年余，泌尿外科会诊不能除外肾盂旁囊肿，现定期复查。反流性食管炎2年，间断口服抑酸药物。发现甲状腺功能减退、甲状腺结节7月余，现口服左甲状腺素钠片50ug qd。发现慢性萎缩性胃炎4月余，间断护胃对症治疗。3月余前考虑抑郁状态，加用氟哌噻吨美利曲辛片*0.5mg/10mg qd至今。40余年前腹部及右前臂大面积烫伤，行植皮手术治疗，目前局部可见疤痕。20年前曾感染“梅毒”，现查TPPA阳性，TRUST阴性，考虑“梅毒转阴”，否认结核、疟疾等传染病病史及其密切接触史。无输血史。否认食物、药物过敏史。

通用模型做结构化（DeepSeek r1）

Thought for 34 seconds

好的，我现在需要处理用户的这个请求，提取病历中患者服用的各种药物及其剂量，并且按照“药品名称：剂量”的格式逐行输出。首先，我需要仔细阅读这篇脱敏的病历，找到所有提到的药物名称和对应的剂量。

首先，用户的需求很明确，是要提取药物和剂量，所以我要确保不要遗漏任何药物信息。同时要注意剂量可能包括不同的单位和用药频率，但用户只需要药品名称和剂量部分，不需要频率。例如“恩替卡韦0.5mg qd”中的“qd”是每天一次的意思，但用户只需要提取0.5mg这部分。

接下来，我逐段分析病历内容：

1. 慢性乙型肝炎部分提到“恩替卡韦0.5mg qd抗病毒治疗至今”，这里药物是恩替卡韦，剂量0.5mg。

2. 2型糖尿病部分，“二甲双胍维格列汀850/50mg bid”，这里需要注意药物名称是复方制剂，剂量是850mg和50mg的组合，可能需要分开列出还是作为复合剂量。但根据病历中的写法，应该作为一个药物名称处理，即二甲双胍维格列汀，剂量是850/50mg。

3. 血管性痴呆部分，“重酒石酸卡巴拉汀1.5mg bid”，药物是重酒石酸卡巴拉汀，剂量1.5mg。

4. 高血压部分有两种药物：“硝苯地平控释片30mg qd”和“厄贝沙坦/氢氯噻嗪片150mg/12.5mg qn”。这里硝苯地平控释片的剂量是30mg，而厄贝沙坦/氢氯噻嗪是复方制剂，剂量分别是150mg和12.5mg，需要合并成一个条目。

5. 甲状腺功能减退部分，“左甲状腺素钠片50ug qd”，剂量50ug。

通用模型做结构化（DeepSeek r1）

- 重酒石酸卡巴拉汀：1.5mg
- 硝苯地平控释片：30mg
- 厄贝沙坦/氢氯噻嗪片：150mg/12.5mg
- 左甲状腺素钠片：50ug
- 氟哌噻吨美利曲辛片：0.5mg/10mg

确认这些是否全部，没有其他药物被提及。例如反流性食管炎的抑酸药物没有具体名称，所以不列出。其他如梅毒治疗已经转阴，没有当前药物。结核、疟疾等也没有用药。因此，以上七个药物是需要提取的。

恩替卡韦：0.5mg
二甲双胍维格列汀：850/50mg
重酒石酸卡巴拉汀：1.5mg
硝苯地平控释片：30mg
厄贝沙坦/氢氯噻嗪片：150mg/12.5mg
左甲状腺素钠片：50ug
氟哌噻吨美利曲辛片：0.5mg/10mg

每次只能提取一类信息

可能有不可预测的额外输出

（注：qd=每日一次，bid=每日两次，qn=每晚一次，但根据要求仅保留剂量信息）

电子病历结构化标准化处理

病历原文

慢性乙型肝炎病史10余年，曾有肝功能异常，中医治疗后好转；1年余前查HBsAg转阴，但肝脏病理提示病毒性肝炎伴肝纤维化（G1S3-4），予恩替卡韦0.5mg qd抗病毒治疗至今。2型糖尿病11年余，目前口服二甲双胍维格列汀850/50mg bid降糖治疗，血糖控制良好。血管性痴呆2年余，长期口服重酒石酸卡巴拉汀1.5mg bid。高血压1年余，血压最高180/80mmHg，长期口服硝苯地平控释片30mg qd，厄贝沙坦/氢氯噻嗪片150mg/12.5mg qn，血压控制在130/80mmHg。发现右侧锁骨下动脉起始部斑块形成1年余，未予特殊诊治。发现左侧肾上腺增粗、左侧肾盂输尿管连接处狭窄、继发左肾积水1年余，泌尿外科会诊不能除外肾盂旁囊肿，现定期复查。反流性食管炎2年，间断口服抑酸药物。发现甲状腺功能减退、甲状腺结节7月余，现口服左甲状腺素钠片50ug qd。发现慢性萎缩性胃炎4月余，间断护胃对症治疗。3月余前考虑抑郁状态，加用氟哌噻吨美利曲辛片*0.5mg/10mg qd至今。40余年前腹部及右前臂大面积烫伤，行植皮手术治疗，目前局部可见疤痕。20年前曾感染“梅毒”，现查TPPA阳性，TRUST阴性，考虑“梅毒转阴”，否认结核、疟疾等传染病病史及其密切接触史。无输血史。否认食物、药物过敏史。


```
{
    "术语": "慢性乙型肝炎",
    "语义类型": "疾病、综合征、病理功能",
    "叙述状态": "存在",
    "身体部位": "无",
    "数值": "NA",
    "单位": "NA",
    "修饰词": "无"
},
{
    "术语": "肝功能异常",
    "语义类型": "症状、体征、临床所见",
    "叙述状态": "存在",
    "身体部位": "无",
    "数值": "NA",
    "单位": "NA",
    "修饰词": "无"
},
{
    "术语": "HBsAg",
    "语义类型": "化学物质、药物",
    "叙述状态": "不存在",
    "身体部位": "NA",
    "数值": "无",
    "单位": "NA",
    "修饰词": "NA"
},
{
    "术语": "肝脏病理",
    "语义类型": "诊断操作",
    "叙述状态": "存在",
    "身体部位": "无",
    "数值": "无",
    "单位": "NA",
    "修饰词": "NA"
},
{
    "术语": "病毒性肝炎",
    "语义类型": "疾病、综合征、病理功能",
    "叙述状态": "存在",
    "身体部位": "无",
    "数值": "NA",
    "单位": "NA",
    "修饰词": "无"
},
{
    "术语": "肝纤维化",
    "语义类型": "疾病、综合征、病理功能",
    "叙述状态": "存在",
    "身体部位": "无",
    "数值": "NA",
    "单位": "NA",
    "修饰词": "无"
},
```

结构化是医疗数据的“测序技术”

- 测序之于基因数据
  - 将生物信息数据化，没有测序技术生物信息就无法获取、数字化或大规模分析，基因研究就停留在理论层面。

- 结构化之于医疗数据
  - 非结构化病历数据难以直接用于分析、共享或决策，结构化的过程将这些数据转化为统一的、机器可读的格式，使其可查询、可分析。没有结构化，医疗数据就像一堆混乱的纸张，无法系统性利用。

- 两者
  - 使数据从原始状态变为可操作知识的桥梁
  - 是不可或缺的基础设施
  - 开启了数据价值的“大门”，推动各自领域的进步

电子病历结构化难在哪里?

- 电子病历是以高度不标准形式写的高度专业的信息
  - 相比常规自然语言处理，更依赖上下位分析和专业背景知识

- 传统系统需要很多处理模块
  - 格式清洗、命名实体识别、词义消歧、实体链接/术语标准化、叙述状态分析、身体部位分析、修饰语分析、数值与单位提取或推断、时间分析……
  - 系统部署对技术要求高，更难以升级；
  - 极少有医院有能力部署全套结构化系统。

- 基于大模型技术，GENIE以一个模型单体代替所有传统模块，并提供更强的分析能力——
  GENIE: Generative Note Information Extraction **病历精灵**
  - https://huggingface.co/THUMedInfo/GENIE_en_8b
  - https://huggingface.co/THUMedInfo/GENIE_zh_7b



电子病历结构化难在哪里?

病历原文

Chest: Resp were unlabored, no accessory muscle use. CTAB, no rales, wheezes or rhonchi.
Abd: Obese, Soft, NTND. No HSM or tenderness.
Ext: Large Right hip hematoma with ecchymoses extending to right flank and down to Right calf. No c/c/edema.
Skin: stasis dermatitis, no ulcers, scars.
Neuro: Alert and oriented x 2, non-focal, CNs II-XII grossly Intact

Pertinent Results:
CTA Chest___: Final Report
REASON FOR EXAM: Desaturation. Clinical concern for pulmonary embolism.

病历实际意思

Chest: Respirations unlabored, no accessory muscle use. Clear to auscultation bilaterally, no rales, wheezes or rhonchi.
Abdomen: Obese, Soft, Non-Tender, Non-Distended. No Hepatosplenomegaly or tenderness.
Extremities: Large Right hip hematoma with ecchymoses extending to right flank and down to Right calf. No clubbing, cyanosis, or edema.
Skin: stasis dermatitis, no ulcers, scars.
Neurological: Alert and oriented to person and place, non-focal, Cranial Nerves II-XII grossly intact.

Pertinent Results:
CTA Chest (Computed Tomography Angiography Chest): Final Report
REASON FOR EXAM: Desaturation. Clinical concern for pulmonary embolism.

GENIE（病历精灵）运行实例

病历原文

Chest: Resp were unlabored, no accessory muscle use. CTAB, no
rales, wheezes or rhonchi.
Abd: Obese, Soft, NTND. No HSM or tenderness.
Ext: Large Right hip hematoma with ecchymoses extending to right
flank and down to Right calf. No c/c/edema.
Skin: stasis dermatitis, no ulcers, scars.
Neuro: Alert and oriented x 2, non-focal, CNs II-XII grossly
Intact

Pertinent Results:
CTA Chest___: Final Report
REASON FOR EXAM: Desaturation. Clinical concern for pulmonary
embolism.


```
{
    "phrase": "auscultation",
    "semantic_type": "Diagnostic Procedure",
    "assertion_status": "present",
    "body_location": "Chest",
    "modifier": "not applicable",
    "value": "null",
    "unit": "not applicable",
    "purpose": "listen to internal body sounds, such as heart and lung sounds"
},
{
    "phrase": "rales",
    "semantic_type": "Sign, Symptom, or Finding",
    "assertion_status": "absent",
    "body_location": "Chest",
    "modifier": "null",
    "value": "not applicable",
    "unit": "not applicable",
    "purpose": "not applicable"
},
{
    "phrase": "wheezes",
    "semantic_type": "Sign, Symptom, or Finding",
    "assertion_status": "absent",
    "body_location": "Chest",
    "modifier": "null",
    "value": "not applicable",
    "unit": "not applicable",
    "purpose": "not applicable"
},
```

GENIE（病历精灵）运行实例

病历原文

Medications on Admission:
Meds: plavix 75mg daily, isosorbide
mononitrate ER 60mg daily,
toprol XL 200mg___, cellcept
1000mg___, nitro 0.4mg prn,
nystatin s/s, prilosec 40mg daily,
ranitidine 300mg qhs,
hydralazine 75mg tid, kayexalate prn,
tacro 9mg___, trazodone
50-100mg qhs, valcyte 450mg daily,
aspirin 325mg daily


```
{
    "phrase": "meds",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "title",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "null",
    "unit": "not applicable",
    "purpose": "null"
},
{
    "phrase": "plavix",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "present",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "75",
    "unit": "mg",
    "purpose": "prevent blood clots"
},
{
    "phrase": "clopidogrel",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "present",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "null",
    "unit": "not applicable",
    "purpose": "prevent blood clots"
},
```

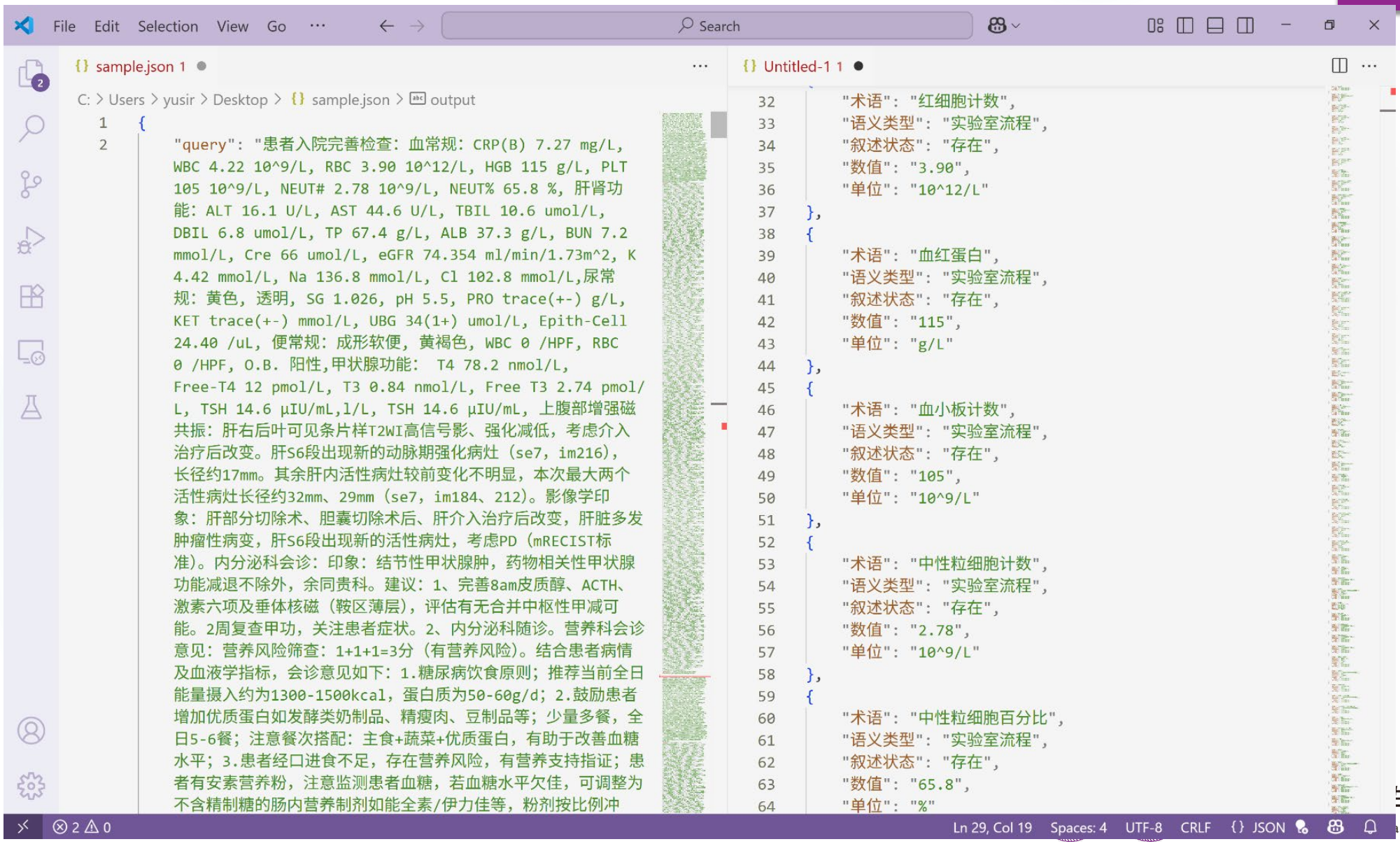
GENIE（病历精灵）运行实例

病历原文

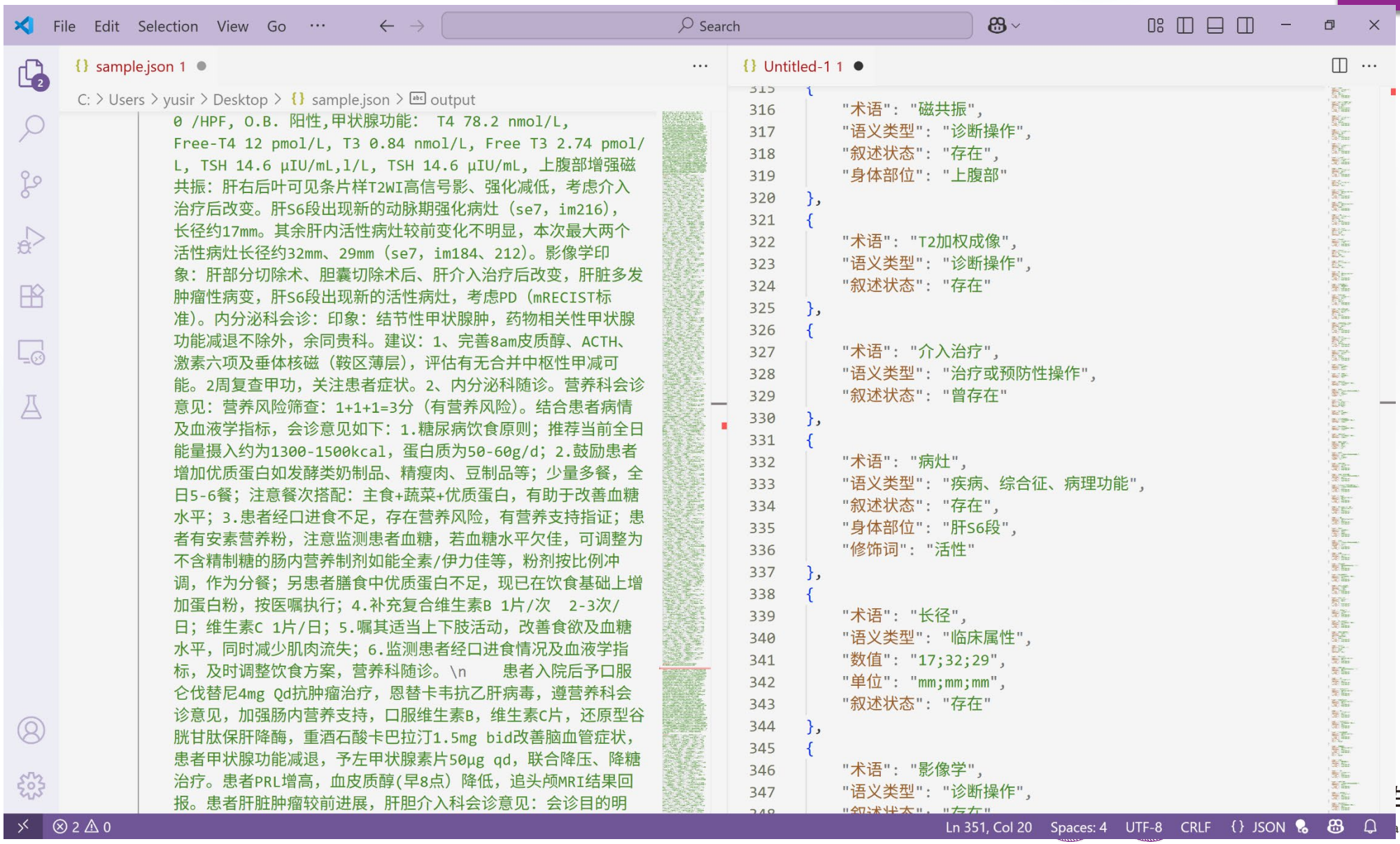
Medications on Admission:
Meds: plavix 75mg daily, isosorbide
mononitrate ER 60mg daily,
toprol XL 200mg___, cellcept
1000mg___, nitro 0.4mg prn,
nystatin s/s, prilosec 40mg daily,
ranitidine 300mg qhs,
hydralazine 75mg tid, kayexalate prn,
tacro 9mg___, trazodone
50-100mg qhs, valcyte 450mg daily,
aspirin 325mg daily


```
{
    "phrase": "isosorbide mononitrate",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "present",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "60",
    "unit": "mg",
    "purpose": "treat angina"
},
{
    "phrase": "toprol xl",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "present",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "200",
    "unit": "mg",
    "purpose": "treat high blood pressure and heart problems"
},
{
    "phrase": "metoprolol succinate",
    "semantic_type": "Chemical or Drug",
    "assertion_status": "present",
    "body_location": "not applicable",
    "modifier": "not applicable",
    "value": "null",
    "unit": "not applicable",
    "purpose": "treat high blood pressure and heart problems"
},
```

GENIE（病历精灵）运行实例



GENIE（病历精灵）运行实例



临床术语系统建设

- 北京协和医院全部电子病历结构化
  - 约五百万术语

- 下一步：数据驱动建立临床标准术语系统
  - 同义词结构
  - 层级结构
  - 标准名称

低频术语示例

sorted_word_dict_with_semantic_100000.txt

```
123888 低密度团片影 （症状、体征、临床所见）：1
123889 弱强化影 （症状、体征、临床所见）：1
123890 额叶占位性病变 （疾病、综合征、病理功能）：1
123891 弥漫性星形细胞瘤 （肿瘤过程）：1
123892 IDH突变型 （肿瘤过程）：1
123893 增强方案化疗 （治疗或预防性操作）：1
123894 影像学稳定 （疾病、综合征、病理功能）：1
123895 弥漫星形细胞瘤 （肿瘤过程）：1
123896 输卵管妊娠物切除 （治疗或预防性操作）：1
123897 妊娠合并右卵巢畸胎瘤 （肿瘤过程）：1
123898 口腔发粘 （症状、体征、临床所见）：1
123899 发闷不适 （症状、体征、临床所见）：1
123900 白细胞计数略偏低 （疾病、综合征、病理功能）：1
123901 面部触觉 （生理功能）：1
123902 空腹血氯 （实验室流程）：1
123903 中度抑郁状态 （精神或行为障碍）：1
123904 胸骨部分坏死 （疾病、综合征、病理功能）：1
123905 胸壁伤口清创缝合术 （治疗或预防性操作）：1
123906 脾脏受压移位 （疾病、综合征、病理功能）：1
123907 肠蠕动波 （症状、体征、临床所见）：1
123908 颈动脉结扎 （治疗或预防性操作）：1
123909 下颌骨部分切除术 （治疗或预防性操作）：1
123910 经动脉重建 （治疗或预防性操作）：1
123911 肌皮瓣重建术 （治疗或预防性操作）：1
123912 多核比例 （实验室流程）：1
123913 TIP方案 （化学物质、药物）：1
123914 ECG提示 （症状、体征、临床所见）：1
123915 大细胞非角化型鳞癌 （肿瘤过程）：1
123916 浆膜面粗糙 （症状、体征、临床所见）：1
123917 下段解剖 （治疗或预防性操作）：1
```

临床术语系统研究进展（潜在同义词搜索）

```
Top 30 neighbors for '多发性硬化':
   1. 多发性硬化 (similarity: 1.0000)
   2. “多发性硬化” (similarity: 1.0000)
   3. 多发生硬化 (similarity: 0.9643)
   4. 多硬化 (similarity: 0.9463)
   5. 多发性经病变 (similarity: 0.9450)
   6. 脑多发硬化病 (similarity: 0.9440)
   7. 多球性硬化 (similarity: 0.9435)
   8. 多发球性硬化 (similarity: 0.9432)
   9. ms (similarity: 0.9412)
  10. 神经多发性硬化 (similarity: 0.9400)
  11. 多发硬化 (similarity: 0.9387)
  12. 多发硬化症 (similarity: 0.9379)
  13. 多发病硬化 (similarity: 0.9358)
  14. MS多发性硬化 (similarity: 0.9358)
  15. 多发性硬化疾病 (similarity: 0.9358)
  16. 多发性硬化症 (similarity: 0.9354)
  17. 神经科多发性硬化 (similarity: 0.9354)
  18. 多发性硬化病 (similarity: 0.9334)
  19. 发性硬化 (similarity: 0.9327)
  20. MS疾病 (similarity: 0.9300)
```

```
Top 30 neighbors for '多囊卵巢综合症':
   1. 多囊卵巢综合症 (similarity: 1.0000)
   2. POCS (similarity: 0.9776)
   3. PCOS (similarity: 0.9696)
   4. PCOS综合征 (similarity: 0.9696)
   5. 多囊卵巢疾病 (similarity: 0.9532)
   6. 多囊卵巢综合症综合征 (similarity: 0.9484)
   7. 多卵巢综合症 (similarity: 0.9453)
   8. 卵巢多囊综合征 (similarity: 0.9422)
   9. P-COS (similarity: 0.9393)
  10. 多囊卵泡综合症 (similarity: 0.9384)
  11. 多巢卵巢综合征 (similarity: 0.9372)
  12. Stein-Leventhal综合征 (similarity: 0.9341)
  13. 多囊卵巢综合征 (similarity: 0.9262)
  14. pcos (similarity: 0.9247)
  15. 囊卵巢综合征 (similarity: 0.9161)
  16. 多囊卵巢综合 (similarity: 0.9161)
  17. 多囊多卵巢综合征 (similarity: 0.9160)
  18. 多囊性卵巢 (similarity: 0.9160)
  19. 多囊卵肿 (similarity: 0.9160)
  20. 子宫多囊卵巢 (similarity: 0.9160)
```

临床术语系统研究进展（潜在同义词搜索）

```
Top 30 neighbors for '代文':
   1. 代文 (similarity: 1.0000)
   2. 代文缬沙坦 (similarity: 0.9649)
   3. 頡纱坦 (similarity: 0.9645)
   4. 缬沙 (similarity: 0.9575)
   5. 缬抄坦 (similarity: 0.9574)
   6. 早缬沙坦 (similarity: 0.9293)
   7. 拜缬沙坦 (similarity: 0.9293)
   8. 缬沙压 (similarity: 0.9293)
   9. 缬纱坦 (similarity: 0.9226)
  10. 缬沙他 (similarity: 0.9102)
  11. 缬继续沙坦 (similarity: 0.9102)
  12. 缬沙抬 (similarity: 0.9102)
  13. 曲缬沙坦 (similarity: 0.9102)
  14. 缬沙坦 (similarity: 0.9102)
  15. 缬赛平 (similarity: 0.9102)
  16. 可缬沙坦 (similarity: 0.9102)
  17. 頡沙坦 (similarity: 0.8770)
  18. 颉沙坦 (similarity: 0.8770)
  19. 缬沙坦纳 (similarity: 0.8770)
  20. 拜厄贝沙坦 (similarity: 0.6646)
```

```
Top 30 neighbors for '神清':
   1. 神清 (similarity: 1.0000)
   2. 神志可清晰 (similarity: 0.9822)
   3. 精神清 (similarity: 0.9822)
   4. 神志请 (similarity: 0.9814)
   5. 神智清晰 (similarity: 0.9803)
   6. 神志神志清 (similarity: 0.9795)
   7. 神清清楚 (similarity: 0.9792)
   8. 神智清 (similarity: 0.9780)
   9. 神志清 (similarity: 0.9778)
  10. 神智清楚 (similarity: 0.9700)
  11. 神志清楚 (similarity: 0.9697)
  12. 意识清 (similarity: 0.9659)
  13. 神志 清晰 (similarity: 0.9651)
  14. 神志清晰 (similarity: 0.9651)
  15. 神智尚清 (similarity: 0.9610)
  16. 神志尚可 (similarity: 0.9602)
  17. 神识清楚 (similarity: 0.9602)
  18. 意识清楚 (similarity: 0.9548)
  19. 精神尚清 (similarity: 0.9517)
  20. 神智尚可 (similarity: 0.9352)
```

```
Top 30 neighbors for '肝阳上亢':
   1. 肝阳上亢 (similarity: 1.0000)
   2. 肝阳升发 (similarity: 0.9835)
   3. 肝阳升动 (similarity: 0.9788)
   4. 肝阳上犯 (similarity: 0.9755)
   5. 肝阳上冲 (similarity: 0.9677)
   6. 肝阳偏亢 (similarity: 0.7368)
   7. 肝阳亢进 (similarity: 0.7258)
   8. 肝气升发 (similarity: 0.7141)
   9. 肝气上冲 (similarity: 0.7012)
  10. 肝阳亢于上 (similarity: 0.7010)
  11. 肝阳亢逆 (similarity: 0.6979)
  12. 肝脏上移 (similarity: 0.6964)
  13. 肝阳偏盛 (similarity: 0.6956)
  14. 风阳升动 (similarity: 0.6947)
  15. 肝阳亢盛 (similarity: 0.6932)
  16. 风阳升而动 (similarity: 0.6883)
  17. 肝上移 (similarity: 0.6803)
  18. 肝阳上扰 (similarity: 0.6677)
  19. 肝阳亢旺 (similarity: 0.6614)
  20. 肝阳有余 (similarity: 0.6591)
```

同义词聚类测试

```
血βhcg水平 (3)  Free-β-HCG (3)  β人绒毛膜促性激素 (2)  B绒毛膜促性腺激素 (2)  检查血hcg (2)
血HCGH (2)  血HCH (2)  β-HCG定量 (2)  β-hCG level (2) 血B-hCG水平 (2)
血B—HCG (2) 血hCG浓度 (1)  监测血BHCG (1)  总β-hCG (1) ββHCG (1)
游离Beta-人绒毛膜粗性腺激素 (1) 血β -HCG (1)  hCG检查 (1) 监测血B-HCG (1) 人绒毛膜促性腺激素的β亚
βCG (1) bHcg (1)  血沉积血HCG (1) βhGC (1)  血HCG指标 (1)
血清βHCG值 (1)  血D—hCG (1) 血β-HCG检查 (1) 血Bhcg (1)  人绒毛膜促性腺激 (1)
BetaHCG (1) 监测血HCG水平 (1)  血的βHCG (1)  HCG -β (1)  定期复查血β-HCG (1)

Cluster 22, Size 75
大便陶土色 (490)  陶土色便 (338)  陶土色大便 (251)  大便呈陶土样 (130)  排白陶土样大便 (122)
大便呈白陶土色 (120)  便色浅 (82) 白陶土样成形便 (33) 灰白陶土样便 (33)  大便颜色变淡 (28)
排便颜色变浅 (22)  大便颜色灰白色 (17) 灰白便 (15) 大便颜色稍浅 (14)  大便白 (14)
大便变为白陶土色 (13)  大便颜色呈陶土样 (13)  粪便颜色变淡 (11)  大便灰色 (9)  大便为陶土色 (8)
大便颜色变浅呈灰白色 (8)  无胆色粪 (7)  白陶色便 (7)  大便呈陶土样便 (7)  大便淡黄色 (6)
大便颜色略浅 (6)  灰白色陶土样大便 (6)  大便色稍浅 (5)  大便陶土样变化 (5)  大便颜色浅白 (4)
灰白样便 (4)  大便呈陶土白色 (4)  陶土样粪便 (4)  大便陶土白色 (4)  大便呈陶土色改变 (4)
大便呈白陶土样软便 (3)  大便颜色较淡 (3)  大便颜色呈变白 (3)  大便颜色略发白 (3)  陶土色粪便 (3)
粪便发白呈陶土色 (3)  大便如白陶土样 (3)  便陶土色 (3)  排陶土色粪便 (2)  大便色浅白 (2)
便颜色变浅呈陶土样 (2)  粪便颜色淡黄白色 (2)  大便颜色变浅色 (2)  大便发白呈陶土样色 (2)  白陶土样
大便呈现白陶土样 (2)  大便瓷白色 (1)  一过性陶土样便 (1)  灰便 (1)  排白陶土色便 (1)
大便颜色为白陶土样 (1)  便发白 (1)  土陶色便 (1)  大便颜色变浅白 (1)  成形白色陶土样大便 (1)
陶样便 (1)  粪便灰白 (1)  大便浅色 (1)  大便黄色变浅 (1)  大便色泽变白 (1)
大便呈白色陶土样样 (1)  偏黄白色大便 (1)  偏白色成形便 (1)  大便陶土颜色 (1)  大便为不成形白陶
便颜色发灰 (1)  陶土洋大便 (1)  浅色大便 (1)  白微黄 (1)  排便陶土色 (1)

Cluster 23, Size 74
可凹陷性水肿 (7130) 凹陷性浮肿 (2248)  可凹陷水肿 (634)  凹性水肿 (347)  压陷性水肿 (99)
凹陷水肿 (70)  可压凹性水肿 (65)  可凹型水肿 (39) 按压凹陷性水肿 (36) 可凹性性水肿 (33)
踝部可凹性水肿 (28) 可凹对称性水肿 (28) 压之凹陷 (21)  手指凹性水肿 (20)  凹陷性性水肿 (16)
```

同义词聚类测试

test_recluster_largest_result_long_output.txt

```
中-重度异形增生 (20)    中到重度不典型增生 (20) 中、重度不典型增生 (17) 细胞中重度异型性 (15)    局灶
中重度异常增生 (15) 中至重度不典型增生 (13) 中度和重度异型增生 (13) 中-重度不典型性增生 (13)    中高
中-重度非典型性增生 (12)    中度-重度不典型增生 (10)    中—重度不典型增生 (9)    上皮中-重度异常增生
腺体呈中-重度异型增生 (8)    中重度不典型性增生 (7)    中-重异型增生 (7)    中度或重度不典型增生 (7)
中-重度异型性增生 (6)    中-高度非典型增生 (6)    中-重度异性增生 (6) 上皮中-高度异型增生 (6) 中重度不
上皮中重度异型增生 (5)    中到重度异型增生 (5)    中及重度不典型增生 (5)    中重度非典型性增生 (4)    粘膜
粘膜中-重度异型增生 (4) 上皮中度至重度异常增生 (4)    中、重度异型增生 (4)    中度-重度异型增生 (3)
细胞中重度异型 (3)    上皮中重度非典型增生 (3)    中-重度细胞异型性 (3)

Cluster 22, Size 3
高级别病变 (4705)    高级别病灶 (5)    高分级病变 (3)

Cluster 23, Size 5
宫颈原位癌 (4335)    宫颈CIS (59)    子宫颈原位癌 (54)    宫颈上皮原位癌 (5)    原位宫颈癌 (4)

Cluster 24, Size 25
中度不典型增生 (3697)    中度异型增生 (1747) 中度非典型增生 (237)    上皮中度异型增生 (139)    上皮内瘤
中度异常增生 (55)    中级别异型增生 (33) 中度异性增生 (32)    中度异型性增生 (23) 中度异形增生 (22)
上皮内病变II级 (20) 上皮内瘤变II级 (13)    上皮中度异常增生 (9)    中度不典型性增生 (9)    中级别上皮内
中度非典型性增生 (7)    不典型增生中度 (6)    中-度不典型增生 (6) 被覆上皮内瘤变II级 (5)    上皮内瘤样变
上皮中度非典型增生 (4)    上皮中度异性增生 (3)    上皮内瘤变II-III级 (3) 上皮中度异型 (3)    上皮内瘤变2级

Cluster 25, Size 13
低级别异型增生 (3478)    低级别瘤变 (379)    低度异型增生 (56)    低度不典型增生 (50) 低级别异形增生
低级别异型性增生 (36)    低级别不典型增生 (31)    低级别异性增生 (27) 低级别异常增生 (11) 低级别非典型
低级别（中度）异形增生 (9)    LGD (6) 低级别异型增生病变 (3)

Cluster 26, Size 5
上皮内瘤变 (3213)    上皮异型增生 (340)    上皮内瘤样变 (67)    上皮内肿瘤 (51) 组织学异型增生 (17)
```

Normal text file length : 66,176 lines : 920 Ln : 1 Col : 1 Pos : 1 Unix (LF) UTF-8 INS